

Deepak Mallampati

APPLIED AI / LLM ENGINEER - HEALTHCARE

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI / LLM Engineer with 4+ years building and shipping production-grade LLM applications in regulated domains. Architect of agentic systems (conductor-subagent orchestration, 42% lower token use) and the RAGAS/BERTScore evaluation harnesses that keep them safe, with deep healthcare experience: production RAG over clinical document corpora (92% precision, ~40% hallucination reduction) and privacy-preserving clinical ML (k-anonymity, l-diversity, differential privacy). Full-stack across Python/TypeScript, FastAPI, and React, with HIPAA/SOC-2 compliance discipline - well suited to owning end-to-end AI applications such as ambient copilots, clinical-trial matching, and automation for clinicians.

CORE COMPETENCIES

End-to-End LLM Applications • Agentic Systems (LangGraph / Multi-Agent) • LLM Evals (RAGAS, BERTScore) • Healthcare RAG & Grounding • HIPAA / SOC-2 Compliance • Inference Optimization • Python / TypeScript Full-Stack

Generative AI & LLMs: GPT-4o/5, Claude, Gemini, LLaMA | LangChain/LangGraph | Agents | RAG | Prompt Engineering | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | Hugging Face

Evaluation & ML: RAGAS, BERTScore, custom domain metrics | PyTorch, TensorFlow, Keras | A/B Testing | Drift Detection | NER, Classification, Summarization

Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Semantic Caching

MLOps & Cloud: AWS (Bedrock, SageMaker, Lambda), Azure OpenAI, GCP Vertex AI | Docker, Kubernetes | CI/CD | MLflow, Weights & Biases | Observability

Security & Compliance: HIPAA, SOC 2, PCI DSS | PII Masking/Redaction | Prompt-Injection Mitigation | Audit Logging

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale **healthcare** document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on clinical queries.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets - K-anonymity, L-diversity, and differential privacy (Laplace) - enabling compliant analytics and model development with zero PII exposure.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Developed **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and supporting **100,000+ requests daily** across 20+ internal teams.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Built supervised regression/classification models for equipment-failure prediction, anomaly detection, and capacity forecasting on operational data.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

Suvidha Foundation

Jan 2022 - Mar 2022 | India

ML Intern

- Built an end-to-end **video summarization** pipeline integrating speech-to-text, transformer abstractive summarization, and timestamp-aligned clip extraction - deployed across **5,000+ videos**, cutting manual editing effort by **70%**.

RESEARCH

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset (k-anonymity $k=3$, l-diversity $l=2$, Laplace differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.
- Designed a composite **privacy-utility scoring framework** benchmarking six anonymization configurations; authored an IEEE-format conference paper documenting methodology and results.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 for controllable long-form generation.

PROJECTS

Privacy-Preserving Clinical ML Pipeline

Framework for analyzing sensitive patient data without exposing identities - k-anonymity, l-diversity, and differential privacy - quantifying exactly how much privacy each technique buys for how little accuracy cost.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across 29+ sites - turning a multi-step manual chore into a single inline action.

career-ops

An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0